

① suma seriei $\sum_{n=1}^{\infty} \frac{2n^2 + n - 2}{(n+1)!}$

$$2n^2 + n - 2 \quad (\Rightarrow) \quad 2n^2 + 2n - n + 1 - 1 \quad (\Rightarrow) \quad \frac{2n(n+1) - (n+1) - 1}{(n+1)!}$$

$$- \text{din (1)} \Rightarrow \frac{2n^2 + n - 2}{(n+1)!} \quad (\Rightarrow) \quad \frac{2n(n+1) - (n+1) - 1}{(n+1)!} \quad (\Rightarrow)$$

$$(\Rightarrow) \frac{2n(n+1)}{(n+1)!} - \frac{n+1}{(n+1)!} - \frac{1}{(n+1)!} \quad (\Rightarrow) \frac{2n(n+1)}{n(n+1)(n-1)!} - \frac{n+1}{(n+1)n!} - \frac{1}{(n+1)!}$$

$$(\Rightarrow) \frac{2}{(n-1)!} - \frac{1}{n!} - \frac{1}{(n+1)!}$$

- notăm $(n-1)!$ ca $k! \Rightarrow \frac{1}{(n-1)!} = \frac{1}{k!} = e$; de aici putem deduce și rezultatele colorate

Suma

- avem

$$\sum_{n=1}^{\infty} \left(\frac{2}{(n-1)!} - \frac{1}{n!} - \frac{1}{(n+1)!} \right) \Rightarrow$$

$$\left\{ \begin{array}{l} \sum_{n=1}^{\infty} \frac{2}{(n-1)!} = 2 \cdot \frac{e}{1} = 2e \\ \sum_{n=1}^{\infty} \frac{1}{n!} = e - 1 \\ \sum_{n=1}^{\infty} \frac{1}{(n+1)!} = e - 2 \end{array} \right.$$

$$\Rightarrow \sum_{n=1}^{\infty} \left(\frac{2}{(n-1)!} - \frac{1}{n!} - \frac{1}{(n+1)!} \right) = 2e - (e - 1) - (e - 2) = 2e - 2e + 3 = \underline{\underline{3}}$$

$$\Rightarrow \left\| \sum_{n=1}^{\infty} \frac{2n^2 + n - 2}{(n+1)!} = 3 \right\|$$

⑥ formula lui Taylor cu rest Lagrange, $n=3$, $a=0$, $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \sqrt[3]{x+1}$$

$$a=0 \Rightarrow \left\{ \begin{array}{l} P_3(x) = \frac{f^{(4)}(c)}{4!} (x)^4 \\ f(x) = f(0) + \frac{f^{(1)}(0)}{1!} x + \frac{f^{(2)}(0)}{2!} x^2 + \frac{f^{(3)}(0)}{3!} x^3 + P_3(x) \end{array} \right.$$

$$\Rightarrow f(x) = f(0) + \frac{f^{(1)}(0)}{1!} x + \frac{f^{(2)}(0)}{2!} x^2 + \frac{f^{(3)}(0)}{3!} x^3 + \frac{f^{(4)}(c)}{4!} x^4$$

$$+ f(0) = \sqrt[3]{1} = 1 \cdot \frac{1}{3} = \underline{\underline{1}}$$

$$\times \frac{f^{(1)}(0)}{1!} = \frac{1}{3} \cdot 1 \left(\frac{1}{3} - 1 \right) = \frac{1}{3} \cdot 1 \left(-\frac{2}{3} \right) = \underline{\underline{\frac{1}{3}}}$$

$$* f^{(2)}(0) = \frac{1}{3} \cdot \left(-\frac{2}{3}\right) \cdot 1^{\left(-\frac{2}{3}-1\right)} = -\frac{2}{9} \cdot 1^{\left(-\frac{5}{3}\right)} = -\frac{2}{9}$$

$$* f^{(3)}(0) = -\frac{2}{9} \cdot \left(-\frac{5}{3}\right) \cdot 1^{\left(-\frac{5}{3}-1\right)} = \frac{10}{27} \cdot 1^{-\frac{8}{3}} = \frac{10}{27}$$

$$* f^{(4)}(c) = \frac{10}{27} \cdot \left(-\frac{8}{3}\right) \cdot (c+1)^{-\frac{11}{3}} = -\frac{80}{81} (c+1)^{-\frac{11}{3}}$$

$$\Rightarrow f(x) = 1 + \frac{\frac{1}{3}}{1!}x + \frac{-\frac{2}{9}}{2!}x^2 + \frac{\frac{10}{27}}{3!}x^3 + R_3(x) \quad (\Rightarrow)$$

$$\boxed{\sqrt[3]{x+1} = 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \frac{5}{81}x^3 + R_3(x)}$$

$$R_3(x) = \frac{f^{(4)}(c)}{4!}x^4 = \frac{-\frac{80}{81}(c+1)^{-\frac{11}{3}}}{24}x^4 = -\frac{10}{243}(c+1)^{-\frac{11}{3}}x^4 \quad (\Rightarrow)$$

$$(\Rightarrow) \boxed{R_3(x) = -\frac{10}{243} \cdot \frac{x^4}{3\sqrt{(c+1)^{11}}}}, \text{ unde } \underline{0 < c < x}$$

② a) demonstrați convergența lui $\sum_{n=1}^{\infty} \frac{1}{n^2}$ folosind Cauchy

$\sum_{n=1}^{\infty}$ - convergență $\forall \varepsilon > 0, \exists n_{\varepsilon} \in \mathbb{N}, \forall n \geq n_{\varepsilon}$ și $m > n_{\varepsilon}, n \leq m, |a_n + \dots + a_m| < \varepsilon$

$$|S_{n+m} - S_n| = \frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \dots + \frac{1}{(n+m)^2} \quad (\Rightarrow)$$

$$* k^2 > k^2 - k \Rightarrow \frac{1}{k^2} < \frac{1}{k^2 - k} \quad (\Rightarrow) \quad \frac{1}{k^2} < \frac{1}{k(k-1)} \quad (1)$$

$$\text{din (1)} \Rightarrow (S_{n+m} - S_n) < \frac{1}{n(n+1)} + \frac{1}{(n+1)(n+2)} + \dots + \frac{1}{(n+m-1)(n+m)} \quad (\Rightarrow)$$

$$(S_{n+m} - S_n) < \left(\frac{1}{n} - \frac{1}{n+1}\right) + \left(\frac{1}{n+1} - \frac{1}{n+2}\right) + \dots + \left(\frac{1}{n+m-1} - \frac{1}{n+m}\right) \quad (\Rightarrow)$$

$$(S_{n+m} - S_n) < \frac{1}{n} - \underbrace{\frac{1}{n+m}}_{>0} \Rightarrow \underline{\underline{(S_{n+m} - S_n) < \frac{1}{n}}}$$

$\Rightarrow$ seria este convergentă, $\forall \varepsilon > 0$

$$b) S = \sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots \Leftrightarrow S = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2}$$

- folosim egalitatea (1) de la a) $\Rightarrow S < 1 + \sum_{n=2}^{\infty} \frac{1}{n(n-1)} \Leftrightarrow$

$$\sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots = 1 - \frac{1}{m}$$

- calculăm limita $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{m} \right)$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{m} \right) = 1 \Rightarrow S < 1 + 1 \Leftrightarrow \boxed{S < 2}$$

③ natura seriei $\sum_{n=1}^{\infty} \frac{1}{3^{1 + \frac{1}{2} + \dots + \frac{1}{n}}}$

- termen general $a_n = \frac{1}{3^{S_n}}$; $S_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}$ - sumă armonică

$$\frac{a_n}{a_{n+1}} = \frac{\frac{1}{3^{S_n}}}{\frac{1}{3^{S_{n+1}}}} = 3^{S_{n+1} - S_n}$$

$$S_{n+1} - S_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} + \frac{1}{n+1} - 1 - \frac{1}{2} - \dots - \frac{1}{n} = \frac{1}{n+1}$$

$$\Rightarrow \frac{a_n}{a_{n+1}} = 3^{\frac{1}{n+1}}$$

- calculăm limita $L = \lim_{n \rightarrow \infty} P_n$; $P_n = n \left(\frac{a_n}{a_{n+1}} - 1 \right)$

$$L = \lim_{n \rightarrow \infty} P_n = \lim_{n \rightarrow \infty} n \cdot \left(3^{\frac{1}{n+1}} - 1 \right) \cdot \frac{1}{n+1} \Leftrightarrow L = \lim_{n \rightarrow \infty} \frac{n}{n+1} \cdot \frac{3^{\frac{1}{n+1}} - 1}{\frac{1}{n+1}}$$

$$* \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$$

$$* \lim_{n \rightarrow \infty} \frac{3^{\frac{1}{n+1}} - 1}{\frac{1}{n+1}} = \lim_{n \rightarrow \infty} \frac{3^k - 1}{k} = \ln 3$$

$$\Rightarrow L = \ln 3 > 1$$

$\Rightarrow$ seria este convergentă

④ convergența seriei $\sum_{n=1}^{\infty} \frac{1}{2^{\sqrt{n}}}$

- termen general $a_n = \frac{1}{2^{\sqrt{n}}}$

$$\ln \frac{1}{a_n} = \ln \frac{1}{\frac{1}{2^{\sqrt{n}}}} \Rightarrow L_n = \frac{\ln \frac{1}{2^{\sqrt{n}}}}{\ln n} \Leftrightarrow L_n = \frac{\ln 2^{\sqrt{n}}}{\ln n} \Leftrightarrow L_n = \frac{\sqrt{n} \ln 2}{\ln n}$$

$$\lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} \frac{\sqrt{n} \ln 2}{\ln n} \Leftrightarrow \ln 2 \cdot \lim_{n \rightarrow \infty} \frac{(\sqrt{n})'}{(\ln n)'} = \ln 2 \cdot \lim_{n \rightarrow \infty} \frac{\frac{1}{2\sqrt{n}}}{\frac{1}{n}} =$$

$$= \ln 2 \cdot \lim_{n \rightarrow \infty} \frac{n}{2\sqrt{n}} = \ln 2 \cdot \infty = \infty \Rightarrow \underline{L_n = \infty} \Rightarrow \boxed{\text{serie este convergentă}}$$

⑨ - serie Taylor în jurul originii; $f: (-1, 1) \rightarrow \mathbb{R}$, $f(x) = \ln(x+1)$
 - origine = 0 $\Rightarrow$ serie Maclaurin $\Rightarrow$ termen general $a_n = \frac{f^{(n)}(0)}{n!}$

$$\ln(x+1) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n, \quad \forall x \in (-1, 1) \Rightarrow$$

$$\Rightarrow \boxed{f(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{(-1)^{n-1}}{n} x^n + \frac{(-1)^n}{n+1} x^{n+1} + \dots}$$

- serie Taylor în jurul originii

⑤ convergența seriei $\sum_{n=1}^{\infty} \frac{\sin n}{n}$

- termeni generali $a_n = \sin n$, $b_n = \frac{1}{n}$; avem doi termeni pentru criteriul lui Dirichlet

* pentru a_n , $S_n = \sum_{k=1}^n \sin k$ - mărginit

$$\sum_{k=1}^n \sin(kx) = \frac{\sin(\frac{nx}{2}) \cdot \sin(\frac{(n+1)x}{2})}{\sin(\frac{x}{2})}; \quad x=1 \Rightarrow S_n = \frac{\sin(\frac{n}{2}) \cdot \sin(\frac{n+1}{2})}{\sin(\frac{1}{2})} \quad (1)$$

$$\sin(x) \in [-1, 1], \quad \forall x \Rightarrow \left| \sin(\frac{n}{2}) \cdot \sin(\frac{n+1}{2}) \right| \leq 1 \quad (2)$$

$$\text{din (1) și (2)} \Rightarrow |S_n| = \frac{\left| \sin(\frac{n}{2}) \cdot \sin(\frac{n+1}{2}) \right|}{\sin(\frac{1}{2})} \leq \frac{1}{\sin(\frac{1}{2})}$$

$\Rightarrow$ condiția de mărginire adevărată constanță M

* pentru b_n , $\sin$ de $|b_n|$ - monoton și descrescător $\forall n$, $n \geq 1$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0, \quad \hookrightarrow \left(\frac{1}{n} > \frac{1}{n+1} \right)$$

$\Rightarrow$ conform Dirichlet, $\boxed{\sum_{n=1}^{\infty} \frac{\sin n}{n} \text{ convergent}}$

$$8) \sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{2n+1}}{2^n (n^3+1)} x^n$$

- termenul general $a_n = (-1)^n \frac{\sqrt{2n+1}}{2^n (n^3+1)}$

a) nota de convergență

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

$$\left| \frac{a_n}{a_{n+1}} \right| = \frac{\frac{\sqrt{2n+1}}{2^n (n^3+1)}}{\frac{\sqrt{2(n+1)+1}}{2^{n+1} ((n+1)^3+1)}} = \frac{\sqrt{2n+1}}{2^n (n^3+1)} \cdot \frac{2^{n+1} ((n+1)^3+1)}{\sqrt{2(n+1)+1}} \Rightarrow$$

$$\Rightarrow \frac{\sqrt{2n+1}}{\sqrt{2(n+1)+1}} \cdot \frac{2^{n+1}}{2^n} \cdot \frac{((n+1)^3+1)}{n^3+1}$$

$$\times \lim_{n \rightarrow \infty} \frac{\sqrt{2n+1}}{\sqrt{2n+3}} = 1$$

$$\times \lim_{n \rightarrow \infty} \frac{2^{n+1}}{2^n} = 2$$

$$\times \lim_{n \rightarrow \infty} \frac{(n+1)^3+1}{n^3+1} = 1$$

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = 1 \cdot 2 \cdot 1 = 2 \Rightarrow \boxed{R=2}$$

b) $x \in (-2, 2) \Rightarrow$ seria este absolut convergentă

$|x| > 2 \Rightarrow$ seria este divergentă

$$\times x = 2 \Rightarrow \sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{2n+1}}{2^n (n^3+1)} \cdot 2^n = \sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{2n+1}}{(n^3+1)}$$

$$\sum_{n=1}^{\infty} \left| (-1)^n \frac{\sqrt{2n+1}}{n^3+1} \right| \Rightarrow \sum_{n=1}^{\infty} \frac{\sqrt{2n+1}}{n^3+1} \approx \frac{\sqrt{2} \cdot n^{\frac{1}{2}}}{n^3} = \frac{\sqrt{2}}{n^{\frac{5}{2}}}$$

- pentru $\sum \frac{1}{n^\alpha}$ și $\alpha > 1$ - seria este convergentă (1)

- din (1) și $\alpha = \frac{5}{2} > 1 \Rightarrow$ seria modulelor este convergentă (2)

- din (1) și (2) $\Rightarrow$ seria convergentă în $x=2$

$$\times x = -2 \Rightarrow \sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{2n+1}}{2^n (n^3+1)} \cdot (-2)^n = \sum_{n=1}^{\infty} (-1)^n \cdot \frac{\sqrt{2n+1}}{(n^3+1)} \cdot (-1) = \sum_{n=1}^{\infty} \frac{\sqrt{2n+1}}{(n^3+1)}$$

- așadar, seria studiată la punctul a) $\Rightarrow$ seria convergentă în $x = -2$

$$(10) \int_0^1 \frac{x}{\sqrt[3]{1-x}} dx$$

a) convergență

$f(x)$ - continuă pe $[0, 1)$; $\lim_{x \rightarrow 1} f(x) = \infty \Rightarrow$ integrală improprie de primă speță

$$f(x) = \frac{x}{\sqrt[3]{1-x}} = \frac{x}{(1-x)^{\frac{1}{3}}}; \quad g(x) = \frac{1}{(1-x)^{\alpha}}; \quad \alpha = \frac{1}{3} \quad (1)$$

$$l = \lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 1} \frac{\frac{x}{(1-x)^{\frac{1}{3}}}}{\frac{1}{(1-x)^{\frac{1}{3}}}} = \lim_{x \rightarrow 1} \frac{x}{1} = \lim_{x \rightarrow 1} x = 1$$

- dacă $\alpha < 1$, $\int_0^1 \frac{1}{(1-x)^{\alpha}} dx$ - convergentă (2)

- din (1) și (2) $\Rightarrow$ integrala este convergentă

$$b) \int_0^1 \frac{x}{\sqrt[3]{1-x}} dx; \quad t^3 = (1-x); \quad dx = -3t^2 dt; \quad x=0 \Rightarrow t=1; \quad x=1 \Rightarrow t=0$$

$$\int_1^0 \frac{1-t^3}{\sqrt[3]{t^3}} \cdot (-3t^2) dt = \int_1^0 \frac{1-t^3}{t} \cdot (-3t^2) dt = -3 \cdot \int_1^0 (1-t^3) \cdot t dt \Leftrightarrow$$

$$\Leftrightarrow 3 \int_0^1 (t - t^4) dt$$

$$= 3 \left[\frac{t^2}{2} - \frac{t^5}{5} \right]_0^1 = 3 \left(\left(\frac{1}{2} - \frac{1}{5} \right) - 0 \right) = 3 \cdot \frac{(5-2)}{10} = \frac{15-6}{10} = \frac{9}{10} \Rightarrow$$

$$\boxed{I = \frac{9}{10}}$$

$$(7) \quad a \in (0, \frac{\pi}{2}), \quad f_n : [0, a] \rightarrow \mathbb{R}, \quad f_n(x) = \sin x + \frac{\sin^2 x}{2} + \dots + \frac{\sin^n x}{n}$$

$$a) \quad f_n(x) = \sum_{k=1}^n u_k(x), \quad u_k(x) = \frac{\sin^k x}{k}; \quad x \in [0, a], \quad a \in (0, \frac{\pi}{2}) \quad (1)$$

$$\left. \begin{array}{l} - \text{avem } 0 \leq \sin x \leq \sin a \\ - \text{din (1)} \end{array} \right\} \Rightarrow \underline{0 \leq \sin x \leq 1};$$

$$q = \sin a; \quad q \in (0, 1) \Rightarrow |u_n(x)| = \left| \frac{\sin^n x}{n} \right| < \frac{q^n}{n}, \quad \forall x \in [0, a]$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{\frac{q^{n+1}}{n+1}}{\frac{q^n}{n}} = \lim_{n \rightarrow \infty} \left(\frac{q^{n+1}}{n+1} \cdot \frac{n}{q^n} \right) = q \cdot \lim_{n \rightarrow \infty} \frac{n}{n+1} = q \cdot 1 = \underline{q}$$

$$\left. \begin{array}{l} |g| < 1 \Rightarrow \sum_{n=1}^{\infty} \frac{g^n}{n} - \text{convergentă} \\ \sum_{n=1}^{\infty} g^n - \text{convergentă} \end{array} \right\} \Rightarrow \boxed{\text{șirul de funcții} \\ \text{uniform convergent}}$$

$$b) f(x) = \lim_{n \rightarrow \infty} f_n(x) = \sum_{n=1}^{\infty} \frac{\sin^n x}{n}$$

$$- \text{folosim Maclaurin} \Rightarrow \ln(1+t) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} t^n, \forall t \in (-1, 1) \Rightarrow$$

$$\ln(1-k) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (-k)^n = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} \cdot (-1)^n}{n} k^n = \sum_{n=1}^{\infty} -\frac{1}{n} k^n = -\sum_{n=1}^{\infty} \frac{k^n}{n}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{k^n}{n} = -\ln(1-k), k \in (-1, 1)$$

$$k = \sin x; x \in [0, a); a < \frac{\pi}{2} \Rightarrow x \in [0, \sin a) \subset [0, 1) \Rightarrow$$

$$\Rightarrow |k| < 1 - \text{condiție de convergență adevărată}$$

$$\Rightarrow \boxed{f(x) = -\ln(1-\sin x)}$$